

GRADE 7 ACCELERATED · UNIT 4

Solving Problems with Rational Numbers

Six lessons in teaching order · the next slide jumps to each one

Contents

4.01	Adding and Subtracting in Scientific Notation	<i>slide 3</i>
4.02–03	Multiplying and Dividing in Scientific Notation	<i>slide 37</i>
4.04	Real-World Problems and Significant Digits	<i>slide 71</i>
4.05	Evaluating Expressions with Radicals	<i>slide 105</i>
4.06	Order of Operations with Radicals	<i>slide 139</i>
4.07–08	Real-World Order of Operations	<i>slide 173</i>

Adding and Subtracting in Scientific Notation

MA.8.NSO.1.5

I can add and subtract two numbers written in scientific notation, and write the answer in scientific notation.

Warm-Up

1. Write 0.00062 in scientific notation.
2. How many times larger is 8×10^7 than 2×10^4 ?
3. Find the value: $\sqrt{144}$
4. Find the sum: $0.49 + 1.2$

Warm-Up

1. Write 0.00062 in scientific notation. **6.2×10^{-4}**
2. How many times larger is 8×10^7 than 2×10^4 ? **4,000**
3. Find the value: $\sqrt{144}$ **12**
4. Find the sum: $0.49 + 1.2$ **1.69**

Notes

I. A power of ten is a place value

10^3	10^4	10^5	10^6
thousands	ten thousands	hundred thousands	millions

$$49,500 + 121,000 = 170,500$$

$$4.95 \times 10^4 + 12.1 \times 10^4 = 17.05 \times 10^4 = 1.705 \times 10^5$$

Two numbers can be added only when they carry the same power of ten.

Notes

II. Same power: add or subtract the coefficients

$$(5.2 \times 10^6) + (3.1 \times 10^6) = 8.3 \times 10^6$$

$$(7 \times 10^{-8}) + (6 \times 10^{-8}) = 13 \times 10^{-8} = 1.3 \times 10^{-7}$$

Add or subtract the coefficients. The power of ten does not change.

Then check the coefficient: if it is not between 1 and 10, the answer is not finished.

Notes

III. Different powers: rewrite one of them first

$$(9.4 \times 10^5) - (8.2 \times 10^3)$$

$$= (9.4 \times 10^5) - (0.082 \times 10^5) = 9.318 \times 10^5$$

$$(1.3 \times 10^3) + (3.4 \times 10^5) = 3.413 \times 10^5, \quad \text{never} \quad 4.7 \times 10^8$$

Rewrite one number so both carry the same power of ten, then add or subtract the coefficients.

Exponent up one, decimal point one place left. The value does not move.

Example 1

$$(8.6 \times 10^7) + (5.9 \times 10^7)$$

Add. Write the answer in scientific notation.

Example 1

$$(8.6 \times 10^7) + (5.9 \times 10^7) = 14.5 \times 10^7$$

$$14.5 \times 10^7 = 1.45 \times 10^8$$

145,000,000

Answer: 1.45×10^8

Your Turn

$$(4.8 \times 10^{-3}) + (7.5 \times 10^{-3})$$

Add. Write the answer in scientific notation.

Your Turn

$$(4.8 \times 10^{-3}) + (7.5 \times 10^{-3})$$

Add. Write the answer in scientific notation.

Answer: 1.23×10^{-2}

Example 2

$$(7.3 \times 10^6) - (4.5 \times 10^5)$$

Subtract. Write the answer in scientific notation.

Example 2

$$4.5 \times 10^5 = 0.45 \times 10^6$$

$$(7.3 \times 10^6) - (0.45 \times 10^6) = 6.85 \times 10^6$$

$$7,300,000 - 450,000 = 6,850,000$$

Answer: 6.85×10^6

Your Turn

$$(5.6 \times 10^{-4}) - (9.0 \times 10^{-6})$$

Subtract. Write the answer in scientific notation.

Your Turn

$$(5.6 \times 10^{-4}) - (9.0 \times 10^{-6})$$

Subtract. Write the answer in scientific notation.

Answer: 5.51×10^{-4}

Whiteboards • Question 1 of 9

$$(3.2 \times 10^5) + (4.1 \times 10^5)$$

Answer it.

Whiteboards • Question 1 of 9

$$(3.2 \times 10^5) + (4.1 \times 10^5)$$

Answer: 7.3×10^5

Whiteboards • Question 2 of 9

A stadium sold 4.2×10^4 tickets on Friday
and 3.5×10^4 on Saturday. Each ticket cost 38 dollars.

How many tickets were sold on the two days?

Answer it.

Whiteboards • Question 2 of 9

A stadium sold 4.2×10^4 tickets on Friday
and 3.5×10^4 on Saturday. Each ticket cost 38 dollars.

How many tickets were sold on the two days?

Answer: 7.7×10^4

Whiteboards • Question 3 of 9

$$(6 \times 10^{-5}) + (7 \times 10^{-5})$$

Answer it.

Whiteboards • Question 3 of 9

$$(6 \times 10^{-5}) + (7 \times 10^{-5})$$

Answer: 1.3×10^{-4}

Whiteboards • Question 4 of 9

$$(5.6 \times 10^8) - (4.9 \times 10^8)$$

Answer it.

Whiteboards • Question 4 of 9

$$(5.6 \times 10^8) - (4.9 \times 10^8)$$

Answer: 7×10^7

Whiteboards • Question 5 of 9

$$(1.3 \times 10^3) + (3.4 \times 10^5)$$

Which is the sum?

A. 4.7×10^5

B. 4.7×10^8

C. 3.413×10^5

D. 3.53×10^5

Answer it.

Whiteboards • Question 5 of 9

$$(1.3 \times 10^3) + (3.4 \times 10^5)$$

Which is the sum?

A. 4.7×10^5

B. 4.7×10^8

C. 3.413×10^5

D. 3.53×10^5

Answer: C — 3.413×10^5

Whiteboards • Question 6 of 9

$$(2.5 \times 10^6) + (9.8 \times 10^6)$$

Which is the sum, written in scientific notation?

A. 12.3×10^6

B. 12.3×10^{12}

C. 1.23×10^6

D. 1.23×10^7

Answer it.

Whiteboards • Question 6 of 9

$$(2.5 \times 10^6) + (9.8 \times 10^6)$$

Which is the sum, written in scientific notation?

A. 12.3×10^6

B. 12.3×10^{12}

C. 1.23×10^6

D. 1.23×10^7

Answer: D — 1.23×10^7

Whiteboards • Question 7 of 9

$$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$$

Answer it.

Whiteboards • Question 7 of 9

$$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$$

Answer: 7.16×10^{-2}

Whiteboards • Question 8 of 9

$$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$$

What is the value of n ?

Answer it.

Whiteboards • Question 8 of 9

$$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$$

What is the value of n ?

Answer: $n = 5.7$

Whiteboards • Question 9 of 9

Devon writes $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$.

Name the two mistakes he made, and give the correct sum.

Write your answer in sentences.

Whiteboards • Question 9 of 9

Devon writes $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$.

Name the two mistakes he made, and give the correct sum.

Answer: 3.413×10^5

Independent Set

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Add and subtract numbers written in scientific notation — HUR

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 2–3

Multiplying and Dividing in Scientific Notation

MA.8.NSO.1.5

I can multiply and divide two numbers written in scientific notation, and write the answer in scientific notation.

Warm-Up

1. $(3.4 \times 10^6) + (2.8 \times 10^6)$
2. Write as one power: $\frac{10^9}{10^4}$
3. Find the value: $\sqrt{169}$
4. Find the quotient: $\frac{4.8}{1.6}$

Warm-Up

1. $(3.4 \times 10^6) + (2.8 \times 10^6)$ **6.2×10^6**

2. Write as one power: $\frac{10^9}{10^4}$ **10^5**

3. Find the value: $\sqrt{169}$ **13**

4. Find the quotient: $\frac{4.8}{1.6}$ **3**

Notes

I. Multiplying: regroup, then use the laws

$$(2 \times 10^4)(3 \times 10^5) = (2 \cdot 3)(10^4 \cdot 10^5) = 6 \times 10^9$$

$$6 \times 10^9, \quad \text{never} \quad 6 \times 10^{20}$$

Multiply the coefficients. Add the exponents — that is the product of powers law, not a new rule.

Notes

II. Dividing: the same move with the quotient law

$$\frac{8.4 \times 10^9}{2.1 \times 10^4} = \frac{8.4}{2.1} \times 10^{9-4} = 4 \times 10^5$$

$$\frac{3 \times 10^8}{3 \times 10^{-4}} = 1 \times 10^{8-(-4)} = 1 \times 10^{12}$$

Divide the coefficients. Subtract the exponents, top minus bottom.

Notes

III. Finish the answer — both directions

$$(8 \times 10^2)(7.5 \times 10^4) = 60 \times 10^6 = 6 \times 10^7$$

$$\frac{2.65 \times 10^{11}}{5 \times 10^{13}} = 0.53 \times 10^{-2} = 5.3 \times 10^{-3}$$

Every answer ends with a coefficient at least 1 and less than 10.

Example 1

$$(4.5 \times 10^{-3})(6 \times 10^7)$$

Multiply. Write the answer in scientific notation.

Example 1

$$(4.5 \times 10^{-3})(6 \times 10^7) = (4.5 \cdot 6)(10^{-3} \cdot 10^7)$$

$$= 27 \times 10^4$$

$$27 \times 10^4 = 2.7 \times 10^5$$

Answer: 2.7×10^5

Your Turn

$$(2.5 \times 10^6)(8 \times 10^{-2})$$

Multiply. Write the answer in scientific notation.

Your Turn

$$(2.5 \times 10^6)(8 \times 10^{-2})$$

Multiply. Write the answer in scientific notation.

Answer: 2×10^5

Example 2

$$\frac{(8 \times 10^2)(7.5 \times 10^4)}{5 \times 10^2}$$

Evaluate. Write the answer in scientific notation.

Example 2

$$(8 \times 10^2)(7.5 \times 10^4) = 60 \times 10^6$$

$$\frac{60 \times 10^6}{5 \times 10^2} = 12 \times 10^4$$

$$12 \times 10^4 = 1.2 \times 10^5$$

Answer: 1.2×10^5

Your Turn

$$\frac{(6 \times 10^5)(4 \times 10^{-3})}{8 \times 10^4}$$

Evaluate. Write the answer in scientific notation.

Your Turn

$$\frac{(6 \times 10^5)(4 \times 10^{-3})}{8 \times 10^4}$$

Evaluate. Write the answer in scientific notation.

Answer: 3×10^{-2}

Whiteboards • Question 1 of 9

$$(3 \times 10^4)(2 \times 10^5)$$

Answer it.

Whiteboards • Question 1 of 9

$$(3 \times 10^4)(2 \times 10^5)$$

Answer: 6×10^9

Whiteboards • Question 2 of 9

A printer makes 4×10^3 copies per hour.

It runs for 2.5×10^2 hours. Paper costs 3 cents a sheet.

How many copies does it make?

Answer it.

Whiteboards • Question 2 of 9

A printer makes 4×10^3 copies per hour.

It runs for 2.5×10^2 hours. Paper costs 3 cents a sheet.

How many copies does it make?

Answer: 1×10^6

Whiteboards • Question 3 of 9

$$\frac{9.6 \times 10^8}{1.2 \times 10^3}$$

Answer it.

Whiteboards • Question 3 of 9

$$\frac{9.6 \times 10^8}{1.2 \times 10^3}$$

Answer: 8×10^5

Whiteboards • Question 4 of 9

$$\frac{2.65 \times 10^{11}}{5 \times 10^{13}}$$

Answer it.

Whiteboards • Question 4 of 9

$$\frac{2.65 \times 10^{11}}{5 \times 10^{13}}$$

Answer: 5.3×10^{-3}

Whiteboards • Question 5 of 9

$$(2 \times 10^4)(3 \times 10^5)$$

Which is the product?

A. 6×10^{-1}

B. 6×10^{20}

C. 6×10^9

D. 5×10^9

Answer it.

Whiteboards • Question 5 of 9

$$(2 \times 10^4)(3 \times 10^5)$$

Which is the product?

A. 6×10^{-1}

B. 6×10^{20}

C. 6×10^9

D. 5×10^9

Answer: C — 6×10^9

Whiteboards • Question 6 of 9

$$(5 \times 10^6)(4 \times 10^3)$$

Which is the product, written in scientific notation?

A. 2×10^9

B. 20×10^9

C. 20×10^{18}

D. 2×10^{10}

Answer it.

Whiteboards • Question 6 of 9

$$(5 \times 10^6)(4 \times 10^3)$$

Which is the product, written in scientific notation?

A. 2×10^9

B. 20×10^9

C. 20×10^{18}

D. 2×10^{10}

Answer: D — 2×10^{10}

Whiteboards • Question 7 of 9

$$\frac{3 \times 10^8}{3 \times 10^{-4}}$$

Answer it.

Whiteboards • Question 7 of 9

$$\frac{3 \times 10^8}{3 \times 10^{-4}}$$

Answer: 1×10^{12}

Whiteboards • Question 8 of 9

$$\frac{8.4 \times 10^7}{n \times 10^2} = 4.2 \times 10^5$$

What is the value of n ?

Answer it.

Whiteboards • Question 8 of 9

$$\frac{8.4 \times 10^7}{n \times 10^2} = 4.2 \times 10^5$$

What is the value of n ?

Answer: $n = 2$

Whiteboards • Question 9 of 9

For $\frac{7.2 \times 10^9}{2.4 \times 10^4}$ Marcus writes 4.8×10^5

and Sara writes 3×10^{13}

Name what each one did wrong, and give the correct quotient.

Write your answer in sentences.

Whiteboards · Question 9 of 9

For $\frac{7.2 \times 10^9}{2.4 \times 10^4}$ Marcus writes 4.8×10^5

and Sara writes 3×10^{13}

Name what each one did wrong, and give the correct quotient.

Answer: 3×10^5

Independent Set

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Multiply numbers written in scientific notation — YZU

Divide numbers written in scientific notation — SGT

Real-World Problems and Significant Digits

MA.8.NSO.1.6

I can solve a real-world problem with numbers in scientific notation and report the answer with the right number of significant digits.

Warm-Up

1. $(4 \times 10^5)(3 \times 10^{-2})$
2. $(6.3 \times 10^7) - (4.1 \times 10^6)$
3. Write 0.00902 in scientific notation.
4. Round 4.738 to the nearest hundredth.

Warm-Up

1. $(4 \times 10^5)(3 \times 10^{-2})$ **1.2×10^4**
2. $(6.3 \times 10^7) - (4.1 \times 10^6)$ **5.89×10^7**
3. Write 0.00902 in scientific notation. **9.02×10^{-3}**
4. Round 4.738 to the nearest hundredth. **4.74**

Notes

I. Which digits are significant

Measurement	Digits	Why
4,207	4	the middle zero counts
0.0058	2	leading zeros never count
3.200	4	trailing zeros count after a decimal point
52,000	2	no decimal point, so they do not

Notes

II. Multiplying or dividing: count the digits

$$(2.5 \times 10^2)(4.13 \times 10^2) = 1.0325 \times 10^5$$

2 significant digits and 3, so the answer keeps 2: 1.0×10^5

Multiply or divide: the answer takes the smaller of the two significant-digit counts.

Notes

III. Adding or subtracting: a different rule

$$1,234 + 5.6 = 1,239.6$$

1,234 stops at the ones place; 5.6 stops at the tenths

so the sum stops at the ones place: $1,240 = 1.240 \times 10^3$

Add or subtract: the answer stops at the leftmost of the two last-digit places.

Example 1

A solar array measures 1.25×10^2 m by 8.6×10^1 m.

What is its area?

Give the answer to the correct number of significant digits.

Example 1

$$(1.25 \times 10^2)(8.6 \times 10^1) = 10,750$$

$1.25 \rightarrow 3$ digits, $8.6 \rightarrow 2$ digits

$$10,750 \rightarrow 11,000 = 1.1 \times 10^4$$

Answer: 1.1×10^4 square meters

Your Turn

A field measures 4.6×10^2 m by 2.35×10^2 m.

What is its area?

Your Turn

A field measures 4.6×10^2 m by 2.35×10^2 m.

What is its area?

Answer: 1.1×10^5 square meters

Example 2

A container holds 3.09×10^{-3} cubic feet of liquid.

3.472×10^{-5} cubic feet evaporates.

How much is left? Give the answer with the correct precision.

Example 2

$$0.00309 - 0.00003472 = 0.00305528$$

$$0.00309 \rightarrow \text{stops at } 10^{-5}$$

$$0.00305528 \rightarrow 0.00306 = 3.06 \times 10^{-3}$$

Answer: 3.06×10^{-3} cubic feet

Your Turn

A road is 3.85×10^3 m long.

A section 2.74×10^2 m long is repaved.

How much of the road is not repaved?

Your Turn

A road is 3.85×10^3 m long.

A section 2.74×10^2 m long is repaved.

How much of the road is not repaved?

Answer: 3.58×10^3 m

Whiteboards • Question 1 of 9

How many significant digits are in 0.00470 ?

Answer it.

Whiteboards • Question 1 of 9

How many significant digits are in 0.00470 ?

Answer: 3

Whiteboards • Question 2 of 9

A pond's surface measures 1.4×10^2 m by 8.5×10^1 m.

The pond is 3 meters deep.

Find the surface area. Round to the correct number of significant digits.

Answer it.

Whiteboards • Question 2 of 9

A pond's surface measures 1.4×10^2 m by 8.5×10^1 m.

The pond is 3 meters deep.

Find the surface area. Round to the correct number of significant digits.

Answer: 1.2×10^4 square meters

Whiteboards • Question 3 of 9

Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^5 .

How should the answer be written?

Answer it.

Whiteboards • Question 3 of 9

Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^5 .

How should the answer be written?

Answer: 7.41×10^5

Whiteboards • Question 4 of 9

$(2.5 \times 10^3) + (4.71 \times 10^2)$ meters, both measured.

Write the sum with the correct precision.

Answer it.

Whiteboards • Question 4 of 9

$(2.5 \times 10^3) + (4.71 \times 10^2)$ meters, both measured.

Write the sum with the correct precision.

Answer: $3.0 \times 10^3 \text{ m}$

Whiteboards · Question 5 of 9

How many significant digits are in 0.03080 ?

A. 4

B. 2

C. 5

D. 3

Answer it.

Whiteboards • Question 5 of 9

How many significant digits are in 0.03080 ?

A. 4

B. 2

C. 5

D. 3

Answer: A — 4

Whiteboards • Question 6 of 9

A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm.

Which is the area, to the correct significant digits?

A. $26.1 \times 10^4 \text{ cm}^2$

B. $2.6 \times 10^5 \text{ cm}^2$

C. $2.6 \times 10^4 \text{ cm}^2$

D. $2.61 \times 10^5 \text{ cm}^2$

Answer it.

Whiteboards • Question 6 of 9

A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm.

Which is the area, to the correct significant digits?

A. $26.1 \times 10^4 \text{ cm}^2$

B. $2.6 \times 10^5 \text{ cm}^2$

C. $2.6 \times 10^4 \text{ cm}^2$

D. $2.61 \times 10^5 \text{ cm}^2$

Answer: B — $2.6 \times 10^5 \text{ cm}^2$

Whiteboards • Question 7 of 9

A car travels 4.20×10^2 km in 3.0 hours.

What is its average speed, to the right number of significant digits?

Answer it.

Whiteboards • Question 7 of 9

A car travels 4.20×10^2 km in 3.0 hours.

What is its average speed, to the right number of significant digits?

Answer: 1.4×10^2 km/h

Whiteboards • Question 8 of 9

A measured length times a measured width gives an area of $3.6 \times 10^3 \text{ m}^2$. The length measured $9.0 \times 10^1 \text{ m}$.

What did the width measure, to two significant digits?

Answer it.

Whiteboards • Question 8 of 9

A measured length times a measured width gives an area of $3.6 \times 10^3 \text{ m}^2$. The length measured $9.0 \times 10^1 \text{ m}$.

What did the width measure, to two significant digits?

Answer: $4.0 \times 10^1 \text{ m}$

Whiteboards • Question 9 of 9

Ellis says: 1,234 has 4 significant digits and 5.6 has 2, so the sum $1,234 + 5.6$ must be written with 2: 1.2×10^3 .

Explain why he is wrong, and give the correct sum.

Write your answer in sentences.

Whiteboards · Question 9 of 9

Ellis says: 1,234 has 4 significant digits and 5.6 has 2, so the sum $1,234 + 5.6$ must be written with 2: 1.2×10^3 .

Explain why he is wrong, and give the correct sum.

Answer: $1,240 = 1.240 \times 10^3$

Independent Set

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Add, subtract, multiply, and divide numbers written in scientific notation: word problems — D5S

Evaluating Expressions with Radicals

MA.8.NSO.1.7

I can evaluate an expression that contains square roots, cube roots and integer exponents.

Warm-Up

1. How many significant digits are in 0.0250 ?
2. $(3 \times 10^4)(2 \times 10^{-7})$
3. Find the value: $\sqrt{196}$
4. Evaluate: $20 - 3 \cdot 4$

Warm-Up

1. How many significant digits are in 0.0250 ? **3**
2. $(3 \times 10^4)(2 \times 10^{-7})$ **6×10^{-3}**
3. Find the value: $\sqrt{196}$ **14**
4. Evaluate: $20 - 3 \cdot 4$ **8**

Notes

I. The radical bar is a grouping symbol

$$\sqrt{25} - \sqrt{36} + \sqrt{1} = 5 - 6 + 1 = 0$$

$$\sqrt{2^3 + 8} = \sqrt{16} = 4$$

$$\sqrt{9 + 16} = \sqrt{25} = 5, \quad \text{never} \quad 3 + 4$$

Everything under the bar is evaluated before the root is taken.

Notes

II. Square roots and cube roots are not the same shape

$$\sqrt{49} = 7$$

$$\sqrt[3]{64} = 4$$

$$\sqrt[3]{-64} = -4$$

$$\sqrt{\frac{1}{9}} = \frac{1}{3}$$

$$\sqrt[3]{\frac{1}{125}} = \frac{1}{5}$$

An odd root keeps the sign of its radicand. A square root sign never gives a negative value.

Notes

III. Radicals alongside exponents

$$5^{-3} = \frac{1}{5^3} = \frac{1}{125} \qquad \sqrt{4^3 - 39} = \sqrt{25} = 5$$

$$\sqrt{4^3 - 39} \cdot 5^{-3} = 5 \cdot \frac{1}{125} = \frac{1}{25}$$

Finish each piece — the radical, the power — and only then put them together.

Example 1

$$\left(-\frac{1}{2}\right)^2 + \sqrt{2^3 + 8}$$

Evaluate.

Example 1

$$\left(-\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\sqrt{2^3 + 8} = \sqrt{16} = 4$$

$$\frac{1}{4} + 4 = \frac{17}{4}$$

Answer: 17/4

Your Turn

$$\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$$

Evaluate.

Your Turn

$$\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$$

Evaluate.

Answer: $-17/9$

Example 2

$$\sqrt[3]{27} - 1.4(\sqrt{3^2 - 5})$$

Evaluate.

Example 2

$$\sqrt[3]{27} = 3, \quad 3^2 - 5 = 4$$

$$\sqrt{4} = 2, \quad 1.4(2) = 2.8$$

$$3 - 2.8 = 0.2$$

Answer: 0.2

Your Turn

$$\sqrt[3]{-8} + 2.5(\sqrt{5^2 - 9})$$

Evaluate.

Your Turn

$$\sqrt[3]{-8} + 2.5(\sqrt{5^2 - 9})$$

Evaluate.

Answer: 8

Whiteboards • Question 1 of 9

$$\sqrt{121}$$

Answer it.

Whiteboards • Question 1 of 9

$$\sqrt{121}$$

Answer: 11

Whiteboards • Question 2 of 9

A square patio has an area of 169 square feet.

Fencing for it costs 12 dollars a foot.

How long is one side of the patio?

Answer it.

Whiteboards • Question 2 of 9

A square patio has an area of 169 square feet.

Fencing for it costs 12 dollars a foot.

How long is one side of the patio?

Answer: 13 feet

Whiteboards • Question 3 of 9

$$\sqrt[3]{-125}$$

Answer it.

Whiteboards • Question 3 of 9

$$\sqrt[3]{-125}$$

Answer: -5

Whiteboards • Question 4 of 9

$$\sqrt{2^3 + 17}$$

Answer it.

Whiteboards • Question 4 of 9

$$\sqrt{2^3 + 17}$$

Answer: 5

Whiteboards • Question 5 of 9

$$\sqrt{9 + 16}$$

Which is the value?

A. 7

B. 25

C. 12.5

D. 5

Answer it.

Whiteboards • Question 5 of 9

$$\sqrt{9 + 16}$$

Which is the value?

A. 7

B. 25

C. 12.5

D. 5

Answer: D — 5

Whiteboards • Question 6 of 9

$$\sqrt[3]{-64}$$

Which is the value?

A. -4

B. not a real number

C. -8

D. 4

Answer it.

Whiteboards • Question 6 of 9

$$\sqrt[3]{-64}$$

Which is the value?

A. -4

B. not a real number

C. -8

D. 4

Answer: A — -4

Whiteboards • Question 7 of 9

$$\sqrt{\frac{1}{9}} + \sqrt[3]{\frac{1}{27}}$$

Answer it.

Whiteboards • Question 7 of 9

$$\sqrt{\frac{1}{9}} + \sqrt[3]{\frac{1}{27}}$$

Answer: $\frac{2}{3}$

Whiteboards • Question 8 of 9

$$\sqrt{4^3 - 39} \cdot 5^{-3}$$

Answer it.

Whiteboards • Question 8 of 9

$$\sqrt{4^3 - 39} \cdot 5^{-3}$$

Answer: 1/25

Whiteboards · Question 9 of 9

Simon says that $\sqrt[3]{-27}$ is not a real number,
because you cannot take the root of a negative.
Explain why he is wrong, and give the value.

Write your answer in sentences.

Whiteboards • Question 9 of 9

Simon says that $\sqrt[3]{-27}$ is not a real number,
because you cannot take the root of a negative.
Explain why he is wrong, and give the value.

Answer: -3

Independent Set

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. This skill continues tomorrow: one assignment, due at the start of the class after 4.06.

Today's skills — all required

Evaluate radical expressions — D8J

Order of Operations with Radicals

MA.8.NSO.1.7

I can evaluate a multi-step expression with rational numbers, exponents and radicals by reading its structure.

Warm-Up

1. $\sqrt{5^2 - 9}$
2. How many significant digits are in 1.020 ?
3. Find the value: $\left(\frac{2}{3}\right)^{-2}$
4. Evaluate: $12 \div 3 \cdot 2$

Warm-Up

1. $\sqrt{5^2 - 9}$ **4**
2. How many significant digits are in 1.020 ? **4**
3. Find the value: $\left(\frac{2}{3}\right)^{-2}$ **9/4**
4. Evaluate: $12 \div 3 \cdot 2$ **8**

Notes

I. The structure, not a mnemonic

	Level	What it means
1	Grouping symbols	(), [], the radical bar, the fraction bar
2	Powers and roots	the same level as each other
3	Multiply and divide	left to right
4	Add and subtract	left to right

Innermost grouping symbol first, then out.

Notes

II. Left to right means left to right

$$12 \div 3 \cdot 2 = 4 \cdot 2 = 8, \quad \text{never} \quad 12 \div 6 = 2$$

$$10 - 4 + 3 = 6 + 3 = 9, \quad \text{never} \quad 10 - 7 = 3$$

Multiplication and division are one level. Addition and subtraction are one level. Left to right, both times.

Notes

III. The fraction bar groups, top and bottom

$$\frac{2^3 - 5}{6^2 - 27} = \frac{3}{9} = \frac{1}{3}$$

$$\frac{2^3 - 5}{6^2} - \sqrt{\frac{4}{9} - \frac{1}{3}} = \frac{1}{12} - \frac{1}{3} = -\frac{1}{4}$$

Finish the numerator and finish the denominator before you divide.

Example 1

$$4(8.63 - (3.2)^2) + \sqrt[3]{125}$$

Evaluate.

Example 1

$$(3.2)^2 = 10.24$$

$$8.63 - 10.24 = -1.61, \quad 4(-1.61) = -6.44$$

$$-6.44 + 5 = -1.44$$

Answer: -1.44

Your Turn

$$3(5.4 - (2.1)^2) + \sqrt{49}$$

Evaluate.

Your Turn

$$3(5.4 - (2.1)^2) + \sqrt{49}$$

Evaluate.

Answer: 9.97

Example 2

$$\frac{2^3 - 5}{6^2} - \sqrt{\frac{4}{9} - \frac{1}{3}}$$

Evaluate.

Example 2

$$\frac{2^3 - 5}{6^2} = \frac{3}{36} = \frac{1}{12}$$

$$\frac{4}{9} - \frac{1}{3} = \frac{1}{9}, \quad \sqrt{\frac{1}{9}} = \frac{1}{3}$$

$$\frac{1}{12} - \frac{1}{3} = -\frac{1}{4}$$

Answer: $-1/4$

Your Turn

$$\frac{3^2 - 1}{4^2} + \sqrt{\frac{7}{16} - \frac{3}{16}}$$

Evaluate.

Your Turn

$$\frac{3^2 - 1}{4^2} + \sqrt{\frac{7}{16} - \frac{3}{16}}$$

Evaluate.

Answer: 1

Whiteboards • Question 1 of 9

$$12 \div 3 \cdot 2$$

Answer it.

Whiteboards · Question 1 of 9

$$12 \div 3 \cdot 2$$

Answer: 8

Whiteboards • Question 2 of 9

A square tile has side $\sqrt{196}$ centimeters.

Its diagonal is about 19.8 centimeters.

What is the perimeter of the tile?

Answer it.

Whiteboards • Question 2 of 9

A square tile has side $\sqrt{196}$ centimeters.

Its diagonal is about 19.8 centimeters.

What is the perimeter of the tile?

Answer: 56 centimeters

Whiteboards • Question 3 of 9

$$10 - 4 + 3$$

Answer it.

Whiteboards · Question 3 of 9

$$10 - 4 + 3$$

Answer: 9

Whiteboards • Question 4 of 9

$$\frac{2^3 - 5}{6^2 - 27}$$

Answer it.

Whiteboards • Question 4 of 9

$$\frac{2^3 - 5}{6^2 - 27}$$

Answer: 1/3

Whiteboards · Question 5 of 9

$$2 + 3 \cdot 4^2$$

Which is the value?

A. 146

B. 50

C. 26

D. 80

Answer it.

Whiteboards · Question 5 of 9

$$2 + 3 \cdot 4^2$$

Which is the value?

A. 146

B. 50

C. 26

D. 80

Answer: B — 50

Whiteboards • Question 6 of 9

$$\frac{6^2 - 4}{2^3}$$

Which is the value?

A. 4

B. 35.5

C. 0.5

D. 256

Answer it.

Whiteboards • Question 6 of 9

$$\frac{6^2 - 4}{2^3}$$

Which is the value?

A. 4

B. 35.5

C. 0.5

D. 256

Answer: A — 4

Whiteboards • Question 7 of 9

$$-2^2 + (-2)^2$$

Answer it.

Whiteboards • Question 7 of 9

$$-2^2 + (-2)^2$$

Answer: 0

Whiteboards • Question 8 of 9

$$12\left(\left(\frac{1}{2}\right)^3 + 3^{-2}\right)$$

Answer it.

Whiteboards • Question 8 of 9

$$12\left(\left(\frac{1}{2}\right)^3 + 3^{-2}\right)$$

Answer: 17/6

Whiteboards • Question 9 of 9

Roscoe says that $(\sqrt{n} - 11)^2$ is always positive,

because a square is always positive.

Is he right? Give a value of n that settles it.

Write your answer in sentences.

Whiteboards • Question 9 of 9

Roscoe says that $(\sqrt{n} - 11)^2$ is always positive,
because a square is always positive.

Is he right? Give a value of n that settles it.

Answer: No — at $n = 121$ the value is 0

Independent Set

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate radical expressions — D8J

Real-World Order of Operations

MA.8.NSO.1.7

I can substitute values into a formula and evaluate it, reading what the situation does rather than hunting for key words.

Warm-Up

1. $-5^2 + (-5)^2$
2. $\sqrt{144} + 2^{-1}$
3. Find the value: $\left(\frac{3}{4}\right)^{-1}$
4. Evaluate $2x^2$ when $x=3$.

Warm-Up

1. $-5^2 + (-5)^2$ **0**
2. $\sqrt{144} + 2^{-1}$ **25/2**
3. Find the value: $\left(\frac{3}{4}\right)^{-1}$ **4/3**
4. Evaluate $2x^2$ when $x=3$. **18**

Notes

I. Substituting: the value goes in parentheses

$$K = \frac{1}{2}mv^2 \quad \text{with} \quad m = 60 \quad \text{and} \quad v = 10:$$

$$\frac{1}{2}(60)(10)^2 = \frac{1}{2}(60)(100) = 3,000 \quad \text{joules}$$

$$\text{with } x = -4: \quad x^2 = (-4)^2 = 16, \quad \text{never } -4^2 = -16$$

Substitute in parentheses, then evaluate by the structure — powers before products.

Notes

II. Key words do not decide the operation

15 dollars plus 4 dollars for each of 6 gigabytes:

$$15 + 4(6) = 39$$

15 dollars for each of 4 lines, plus 6 dollars of tax:

$$15(4) + 6 = 66$$

The same words, the same numbers, different structures. Read the situation.

Notes

III. Round once, at the end — and say the unit

$$6\sqrt{200} = 84.8528... \approx 84.85 \text{ feet}$$

rounding the side to 14 first: $6(14) = 84 \text{ feet}$

Carry the exact value, round once at the end, and write the unit.

Example 1

Kinetic energy is $K = \frac{1}{2}mv^2$,

for mass m in kilograms and speed v in meters per second.

Find K for a car of mass 1,200 kg travelling at 15 m/s.

Give the answer with its unit.

Example 1

$$\begin{aligned} K &= \frac{1}{2}(1,200)(15)^2 \\ &= \frac{1}{2}(1,200)(225) \\ &= 600(225) = 135,000 \end{aligned}$$

Answer: 135,000 joules

Your Turn

Find $K = \frac{1}{2}mv^2$ for $m = 850$ kg and $v = 12$ m/s.

Your Turn

Find $K = \frac{1}{2}mv^2$ for $m = 850$ kg and $v = 12$ m/s.

Answer: 61,200 joules

Example 2

A square floor of side 9 feet is carpeted.

Carpet is 2.89 dollars a square foot and padding is 0.75 dollars a square foot.

Installation is a flat 95 dollars. Find the total cost.

Example 2

$$\begin{aligned} &9^2(2.89 + 0.75) + 95 \\ &= 81(3.64) + 95 \\ &= 294.84 + 95 = 389.84 \end{aligned}$$

Answer: 389.84 dollars

Your Turn

A square bathroom floor of side 6 feet is tiled.

Tile is 4.25 dollars a square foot and grout is a flat 13.48 dollars.

Find the total cost.

Your Turn

A square bathroom floor of side 6 feet is tiled.

Tile is 4.25 dollars a square foot and grout is a flat 13.48 dollars.

Find the total cost.

Answer: 166.48 dollars

Whiteboards • Question 1 of 9

Evaluate $3x^2$ when $x = -2$.

Answer it.

Whiteboards • Question 1 of 9

Evaluate $3x^2$ when $x = -2$.

Answer: 12

Whiteboards · Question 2 of 9

A cyclist has a mass of 70 kilograms and rides at 8 meters per second.

Her bicycle cost 480 dollars.

Kinetic energy is $K = \frac{1}{2}mv^2$, for a mass m and a speed v .

Find her kinetic energy.

Answer it.

Whiteboards · Question 2 of 9

A cyclist has a mass of 70 kilograms and rides at 8 meters per second.

Her bicycle cost 480 dollars.

Kinetic energy is $K = \frac{1}{2}mv^2$, for a mass m and a speed v .

Find her kinetic energy.

Answer: 2,240 joules

Whiteboards • Question 3 of 9

A plan costs 15 dollars plus 4 dollars for each gigabyte.
Find the cost of 6 gigabytes.

Answer it.

Whiteboards • Question 3 of 9

A plan costs 15 dollars plus 4 dollars for each gigabyte.
Find the cost of 6 gigabytes.

Answer: 39 dollars

Whiteboards • Question 4 of 9

A square floor of side 8 feet is tiled at 3.50 dollars a square foot.

Installation is a flat 60 dollars.

Find the total cost.

Answer it.

Whiteboards • Question 4 of 9

A square floor of side 8 feet is tiled at 3.50 dollars a square foot.

Installation is a flat 60 dollars.

Find the total cost.

Answer: 284 dollars

Whiteboards • Question 5 of 9

Evaluate $\frac{1}{2}mv^2$ when $m = 40$ and $v = 3$.

Which is the value?

A. 180

B. 120

C. 360

D. 3,600

Answer it.

Whiteboards • Question 5 of 9

Evaluate $\frac{1}{2}mv^2$ when $m = 40$ and $v = 3$.

Which is the value?

A. 180

B. 120

C. 360

D. 3,600

Answer: A — 180

Whiteboards • Question 6 of 9

A square patio of side 12 feet is sealed at 1.25 dollars a square foot.

Delivery of the sealer is a flat 38 dollars.

Which expression gives the total cost?

A. $12(1.25) + 38$

B. $12^2 + 1.25 + 38$

C. $12^2(1.25 + 38)$

D. $12^2(1.25) + 38$

Answer it.

Whiteboards • Question 6 of 9

A square patio of side 12 feet is sealed at 1.25 dollars a square foot.

Delivery of the sealer is a flat 38 dollars.

Which expression gives the total cost?

A. $12(1.25) + 38$

B. $12^2 + 1.25 + 38$

C. $12^2(1.25 + 38)$

D. $12^2(1.25) + 38$

Answer: D — $12^2(1.25) + 38$

Whiteboards • Question 7 of 9

A fence runs along 6 sides, each $\sqrt{200}$ feet long.

Find its total length, to the nearest hundredth of a foot.

Answer it.

Whiteboards • Question 7 of 9

A fence runs along 6 sides, each $\sqrt{200}$ feet long.

Find its total length, to the nearest hundredth of a foot.

Answer: 84.85 feet

Whiteboards • Question 8 of 9

Evaluate $\frac{a^2 - b}{c}$ when $a = 7$, $b = 4$, $c = 5$.

Answer it.

Whiteboards • Question 8 of 9

Evaluate $\frac{a^2 - b}{c}$ when $a = 7$, $b = 4$, $c = 5$.

Answer: 9

Whiteboards • Question 9 of 9

A plan costs 15 dollars plus 4 dollars for each of 6 gigabytes.

Another costs 15 dollars for each of 4 lines, plus 6 dollars of tax.

Find both totals, and say why the key words could not have told you.

Write your answer in sentences.

Whiteboards • Question 9 of 9

A plan costs 15 dollars plus 4 dollars for each of 6 gigabytes.

Another costs 15 dollars for each of 4 lines, plus 6 dollars of tax.

Find both totals, and say why the key words could not have told you.

Answer: 39 dollars and 66 dollars

Independent Set

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate variable expressions: word problems — DPG